THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Silver articles turn black on exposure to air because of the formation of: 1

(a) silver oxide **(b)** silver sulphide **(c)** silver chloride **(d)** silver nitrate

2. When an iron nail is dipped in copper sulphate solution, the colour of the solution changes from: 1

(a) blue to light green **(b)** green to blue
(c) blue to colourless **(d)** colourless to blue

3. Respiration is an example of an: 1

(a) precipitation reaction **(b)** exothermic reaction
(c) displacement reaction **(d)** endothermic reaction

4. Copper articles acquire a green coating of: 1

(a) copper sulphate **(b)** copper chloride
(c) basic copper carbonate **(d)** copper oxide

5. White silver chloride turns grey in sunlight because: 1

(a) it dissolves in moisture
(b) it decomposes to silver metal and chlorine
(c) it combines with oxygen
(d) it gets reduced to silver oxide

6. Photosynthesis is: 1

(a) a neutralisation reaction **(b)** an endothermic reaction
(c) a displacement reaction **(d)** an exothermic reaction

7. A balanced chemical equation follows the law of: 1

(a) conservation of energy only **(b)** conservation of mass
(c) multiple proportions **(d)** constant proportions

8. The reaction of quick lime with water is: 1

(a) displacement and endothermic **(b)** double displacement and exothermic
(c) decomposition and endothermic **(d)** combination and exothermic

9. Brown fumes are given off on heating lead nitrate. The gas is: 1

(a) oxygen **(b)** carbon dioxide **(c)** nitrogen dioxide **(d)** sulphur dioxide

10. In the reaction $\mathrm{Fe} + \mathrm{CuSO_4} \longrightarrow \mathrm{FeSO_4} + \mathrm{Cu}$, the metal displaced is: 1

(a) sulphur **(b)** oxygen **(c)** copper **(d)** iron

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a precipitation reaction? Give one example. 2

12. Why is it necessary to balance a chemical equation? 2

13. Why is the decomposition reaction of calcium carbonate called a thermal decomposition? Write the equation. 2

14. Write the balanced equation for sodium reacting with water to form sodium hydroxide and hydrogen. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Write balanced chemical equations for: 3

(a) calcium hydroxide + carbon dioxide,
(b) zinc + silver nitrate,
(c) aluminium + copper chloride.

16. What is rancidity? State its effect and two methods of prevention. 3

17. Why is photosynthesis an endothermic and respiration an exothermic reaction? Give the equation of respiration. 3

OR

Lead nitrate is heated in a boiling tube. State the observation, write the equation and name the type of reaction. 3

18. What is corrosion? Give two examples and one harmful effect. 3

OR

Classify as exothermic or endothermic: 3

(a) burning of coal
(b) decomposition of vegetable matter into compost
(c) action of sunlight on silver bromide.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. What is a displacement reaction? Describe the iron nail and copper sulphate activity with observations, equation and inference. 5

OR

Describe the electrolysis of water in Activity 1.7. Name the gases, their ratio, and tests. 5

20. Explain double displacement reaction with the example of sodium sulphate and barium chloride. Also give another example with an observation. 5

OR

What is a combination reaction? Describe the reaction of quick lime with water and give two other examples. 5

END OF QUESTION PAPER Total: 40 Marks